



Challenges in data documentation and retrieval in the ASHA diary from a usability perspective

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ABSTRACT

In India, Accredited Social Health Activists (ASHA) serve as indispensable connectors between healthcare systems and underserved communities, ensuring the delivery of essential health services. At the core of their responsibilities lies the meticulous documentation of patient information in ASHA diaries. ASHA diaries are essential tools for documenting and retrieving accurate information, enabling the monitoring of health status, tracking of interventions, and supporting data-driven decision-making. However, navigating and extracting relevant information from ASHA diaries presents significant challenges for ASHA workers. Therefore, the primary objective of this research is to assess the usability of the current ASHA diary by identifying the key challenges ASHA workers encounter in documenting and retrieving effective data. An empirical study encompassing both qualitative and quantitative methods was carried out involving fifty-seven ASHA workers. Quantitative data was collected through a modified questionnaire. Qualitative data were gathered via observation, informal interviews, and task analysis to better understand the difficulties in using the diary. Descriptive statistics and factor analyses were conducted to analyze and describe the factors contributing to usability issues. The findings reveal various challenges, including poor visual cues (visibility, readability, legibility), difficulty in comprehension, inconsistent data presentation, excessive cognitive load, inappropriate resource utilization, etc. By identifying these usability challenges, this research provides substantial insights for improving the design of ASHA diaries, ultimately enhancing data documentation and retrieval processes for ASHA workers. This research makes a meaningful contribution to optimizing healthcare information systems to better meet the needs of ASHA workers and improve healthcare delivery in marginalized communities.

1. Introduction

The National Rural Health Mission (NRHM) was launched by the Government of India (GoI) in 2005 to strengthen the public healthcare system, particularly in rural areas, to achieve the Millennium Development Goals (MDGs). The primary objective of NRHM is to provide accessible, accountable, affordable, effective, and reliable primary healthcare, addressing various health determinants such as maternal and childcare, nutrition, sanitation, etc. Under the mission, a network of female health volunteers known as “Accredited Social Health Activists” (ASHA) was introduced for each village across the country to serve as a vital link between underserved communities and healthcare systems (Das and Dasgupta, 2015; Kumar et al., 2013; Panda et al., 2019; Sugandha and Jagannath, 2019). Over time, the ASHA program has globally become the largest community health worker (CHW) initiative (Ministry of External Affairs. Government of India, 2022; Tripathy et al.,

2016). ASHA workers are selected from the same community or village they serve. One ASHA worker, preferably within the age group of 25–45 years, is assigned for every one-thousand population. The minimum educational qualification for an ASHA worker is set at the 8th standard, but this requirement is relaxed to the 5th standard if a suitable candidate is unavailable. The primary responsibilities of ASHA workers include health education, escorting patients to hospitals, diagnosing and counselling, community mobilization, providing essential medicines, raising community awareness, and conducting health surveys (Gopalan et al., 2012; Shet et al., 2018; Shrivastava et al., 2023). National evaluation surveys have shown gradual improvements in maternal and child health (MCH) following the implementation of the ASHA program (Kaur et al., 2022; Shilpa et al., 2020).

There is a global shortage of health workers (HW), especially in the South-East Asia Region (World Health Organization, 2018). As of 2020, approximately 9,14,101 ASHA workers were employed, falling short of

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the target of 9,56,672 workers (Press Information Bureau. Government of India, 2020; Ministry of Health and Family Welfare. Government of India, 2021). Shihundla et al. (2016) highlighted that CHWs experience heavy workloads in healthcare systems due to their positioning at the sharp end of the system. According to ASHA operational guidelines, their standard working hours were expected to be 16 h per week or 2–3 h daily (Agarwal et al., 2019). However, the scope of the program has expanded over time, including additional tasks, surveys, and programmes. Currently, the ASHA program includes more than 30 tasks, some of which are unrelated to health and may take precedence over health-related responsibilities (Manjunath et al., 2022; Shelley et al., 2018; Smith et al., 2014; Tripathy et al., 2016). Eventually, the growing responsibilities have extended the duty limit beyond 8 h per day, leaving minimal personal time (Abdel-All et al., 2019). This excessive workload has further contributed to significant stress, fatigue, and burnout among health workers (Pulagam and Satyanarayana, 2021; Gopalan et al., 2012).

Research indicates that burdening HWs with numerous responsibilities, leading to role confusion, difficulty in prioritizing tasks, and reducing service quality (Manjunath et al., 2022; Shelley et al., 2018; Smith et al., 2014). Additionally, an increased workload can result in mismanagement of the information, leading to incomplete patient records or inaccurate documentation, which may cause the omission of critical data (Mondal and Murhekar, 2018; Mutshatshi et al., 2018; Shihundla et al., 2016).

Documentation is essential for tracking the activities and services provided by ASHA workers for patient treatment (Marinić, 2015; Oztekin et al., 2013). It plays a key role in maintaining records of health interventions, service delivery, and community engagement, which are vital for evaluating the program's effectiveness and functionality (Ministry of Health and Family Welfare. Government of India, 2013; World Health Organization, 2009). However, various studies highlighted that the exchange of accurate information is challenging, especially in rural areas where timely communication is essential for managing community health conditions (Aceto et al., 2018; Jackson et al., 2017; Sokey et al., 2018). O'Mahony et al. (2014) and Verdezoto et al. (2021) discussed the challenges HWs face in collecting, managing, and reporting quality health data across the country. Globally, there is a growing emphasis on improving the quality of health workers' documentation (Khattak et al., 2016; Nakate et al., 2022; Okaisu et al., 2016; World Health Organization, 2009). Gencturk et al. (2017), Kruk et al. (2018), and Mutshatshi et al. (2018) highlighted that maintaining detailed records facilitates effective communication within and across healthcare systems, ensuring quality care.

The literature underscores the importance of maintaining high-quality health records (Kruk et al., 2018; Shihundla et al., 2016). ASHA workers are responsible for documenting various health records and assisting in monitoring the health status of the community through the ASHA diary, which is an official paper record (PPR). Muinga et al. (2021) noted that PPRs are pivotal in facilitating multidisciplinary team care and tracking patient progress. Despite the increasing adoption of electronic patient records (EPRs), PPRs remain a dominant medium for recording patient care, especially in low- and middle-income countries (LMICs). However, the lack of research on how PPR development hampers efforts to improve the designs and their quality. It is important to note that many existing records are outdated and inefficient, affecting patient care. Bist et al. (2023) and Kohli et al. (2015) reported that ASHA workers often struggle with understanding the diary, filling out extensive information, and needing sufficient time to complete the ASHA diary. These challenges result in errors, inaccuracies, incomplete records, unnecessary burdens in interpreting information, poor decision-making, delays in information delivery, and reduced patient care time (Ash et al., 2004; Kim et al., 2017; Pool et al., 2024). Consequently, these inefficiencies in the documentation process impact the performance of ASHA workers and the overall quality of healthcare delivery in rural areas.

Thus, based on the literature, the usability of PPRs remains a critical factor that influences the efficiency and effectiveness of healthcare delivery. Effective usability of PPRs ensures that healthcare providers can easily access, record, and retrieve patient information, directly impacting the quality of care. The usability of PPRs is essential for facilitating effective communication, reducing errors, saving time, and preparing the groundwork for future digital systems (Emmerich, 2023; Muinga et al., 2021; Pfeuffer et al., 2022).

Addressing these issues, the ability to track health information over time is invaluable in rural settings, where resources are often limited, and formal health infrastructure may be sparse. By maintaining detailed records, ASHA workers can identify at-risk individuals early, ensuring timely interventions and improving health outcomes. Accurate and up-to-date records enable ASHA workers to provide essential information to higher-level healthcare providers, such as doctors and nurses, who may be based in distant health centers. Thus, usability is crucial not only for enhancing the speed, accuracy, and quality of tasks performed by workers but also for ensuring the success of the documentation system. To the best of our knowledge, the usability of the ASHA diary has not been evaluated yet. Therefore, the primary objective of the current research is to conduct a comprehensive analysis to assess the usability of the ASHA diary by identifying the key challenges ASHA workers face in documenting and retrieving information. The results of this study will aid in designing more effective and efficient document systems for ASHA workers, improving their usability, and helping maintain accurate health records within rural health systems.

2. Methods

This research employed an empirical research design to investigate the usability challenges ASHA workers face in documenting and retrieving rural health information using the ASHA diary. A mixed-method approach was used for data collection, with a detailed explanation of participant selection. The data collection methods were designed to thoroughly analyze usability issues by integrating relevant usability principles.

2.1. Study design

An empirical study was conducted using a mixed-method approach with a combination of quantitative and qualitative analysis between April and July in Jabalpur, India. The quantitative component was analyzed, simultaneously with qualitative analysis, for an in-depth understanding of the factors affecting ASHA workers. To maintain their diary, a data collection tool is provided yearly by the government in the months of March and April every year. A set of modified questionnaires (kindly refer to Appendix A) was used to collect data, followed by the conduction of interviews, task analysis, and observational study to understand the pattern of the problem more deeply (Bist et al., 2023; De Groot et al., 2022; Khairat et al., 2019; Koopmann et al., 2023; Meena et al., 2020).

2.2. Participants

The investigation employed the multistage sampling design recommended by the National Health Systems Resource Center (NHSRC), New Delhi, to evaluate the ASHA program in India (Mahajan and Kaur, 2022). The sampling process involved several stages: selecting a district, then blocks, and finally villages (see Fig. 1). The district was chosen purposively sampling technique based on the data from the Ministry of Health and Family Welfare (MoHFW), focusing on areas where women had fewer than four antenatal care visits and had become mothers between the ages of 15–19 years, with early marriages, i.e., before 18 years (referred to as “NHM-MP”). Within the selected district, two underperforming blocks were identified and included in the study through purposive sampling. Four primary health centers (PHCs) were randomly

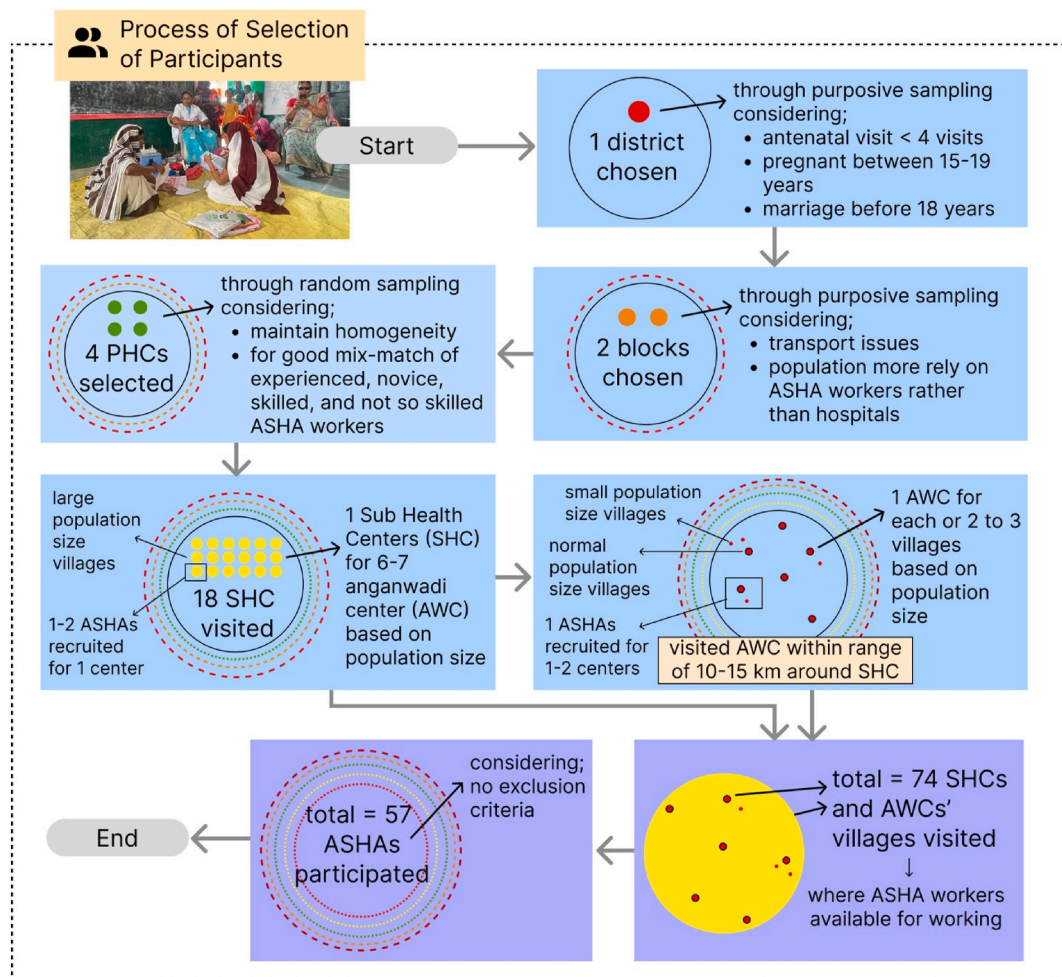


Fig. 1. Process of selection of participants.

selected from the two underperforming blocks based on the research convenience. Subsequently, eighteen sub-health centers (SHCs) were then chosen along with nearby Anganwadi centers (AWCs) within a range of 10–15 km. This process resulted in the selection of approximately seventy-four villages for the current research. All ASHA workers from the selected villages, including those managing 2–3 villages, were invited to participate, and all 57 ASHA workers agreed. Participation was voluntary with no exclusion criteria (Kohli et al., 2015; Shihundla et al., 2016).

2.3. Data collection

In quantitative research methods, a modified set of usability questionnaire was developed, expecting to comprehensively cover all major aspects of usability constructs in a detailed and unified manner. Initially, the questions incorporated several well-validated usability frameworks identified from existing models and standards in the literature (Bevan, 1995; Brooke, 1996; Koopmann et al., 2023; Lee and Kozar, 2012; Lund, 2001; Murphy et al., 2019; Oztekin et al., 2013; Seffah et al., 2006). Subsequently, the constructs were obtained and refined for relevance, applicability, and convenience, incorporating overlapping constructs from modified questionnaires. To address the diverse usability criteria of the ASHA diary, the research adopted 11 relevant usability criteria and included 37 specific metric items. A 5-point Likert scale ranging from “Strongly disagree” to “Strongly agree”, with a neutral category of “Neither agree nor disagree”, was used to gather appropriate responses (Brooke, 1996; Gopalan et al., 2012; WHO, 2019). The questionnaire included closed-ended questions to record the socio-demographic

profile of ASHA workers (Kohli et al., 2015; Panda et al., 2019). The questionnaire, initially in English and later translated into the local language for accurate data collection, was pilot-tested on 10 subjects. Based on their feedback, the instrument was finalized with improvements in language structure, clarity, and spelling. A total of 45 questions were used to determine the demographic profile and usability factors affecting ASHA workers in documenting and retrieving information from the ASHA Diary.

In the qualitative research method, a proficient female researcher conducted observations, informal interviews, and task analyses to understand the challenges ASHA workers encountered in documenting and retrieving information. Being fluent in the local language helps the researcher to collect data effectively by creating an environment where they feel comfortable and can discuss their concerns. The observations were conducted in real-world settings, such as rural health clinics and community centers (SHC and AWC), where ASHA workers typically perform their tasks. The observation focused on management burden, environmental effects, retrieval difficulties, and cognitive loads. Face-to-face informal interviews were conducted at convenient locations, such as residences and AWC, with follow-up questions posed based on the depth of discussion and participant engagement. The informal interview focused on usability concerns and system understanding. Additionally, two to three diverse tasks were administered to thoroughly dissect challenges such as time completion, navigation difficulties, and execution delays in documenting and retrieving information. The process spanned 70–75 min, during which the investigator diligently recorded the data using field notes and, where permissible, audio-video recordings to capture detailed interactions and usability issues (Murphy

et al., 2019; Shihundla et al., 2016; Bist et al., 2023). This integrated Quanti-Quali Triangulation (see Fig. 2) design enabled both statistical validation and contextual interpretation of usability challenges faced by ASHA workers, enhancing the robustness and applicability of the findings (Jick, 1979).

2.4. Measurement

The data was collected to examine the usability problem of ASHA workers in documenting and retrieving information from the ASHA Diary. For evaluation, we defined the assessment areas based on usability principles that overlapped with the research as follows: Visual cues (VC) are defined as readability, legibility, and visibility being easily understood and conveniently done for routine workflow (Lee and Kozar, 2012; Oztekin et al., 2013; Seffah et al., 2006). Content relevance (CR) is defined as the extent to which the content is pertinent and up-to-date (Lee and Kozar, 2012). Comprehensibility/Learnability (CL) is defined in the context of ASHA workers, who can understand the intended meaning, easily learn the functionality and gain proficiency to complete the tasks (Koopmann et al., 2023; Lee and Kozar, 2012). Simplicity (SL) is defined as the extraneous elements that are eliminated from the user interface without significant loss of information (Lee and Kozar, 2012; Seffah et al., 2006). Consistency (Co) is defined as the degree of uniformity among elements of the user interface and whether they offer meaningful metaphors to users (Lee and Kozar, 2012; Oztekin et al., 2013; Seffah et al., 2006). Findability/Navigability (FN) means whether users can move around in the interface efficiently and locate the relevant content (Lee and Kozar, 2012; Seffah et al., 2006). Information prioritization (IP) refers to organizing information based on its relevance and frequency of use, ensuring it is easily accessible and appropriately positioned (Murphy et al., 2019). Error prevention (EP) is defined as

anticipating potential mistakes and incorporating features to guide, restrict, and inform users in a way that minimizes the chance of errors and their impact (Bevan et al., 2015; Koopmann et al., 2023; Oztekin et al., 2013). Minimal memory load (MM) is defined as whether a user is required to keep a minimal amount of information in mind to achieve a specified task (Murphy et al., 2019; Oztekin et al., 2013). Resources utilization (RU) is defined as if the diary can consume appropriate amounts and types of resources for a given timeline (Bevan et al., 2015; Seffah et al., 2006). Portability (Po) is defined as whether the tools can be transferred from one environment to another (Bevan et al., 2015).

2.5. Data analysis plan

The main purpose of this research was to explore the dimensions of usability that influence the ease and efficiency with which ASHA workers complete their diaries. In this study, exploratory factor analysis (EFA), a statistical method, was employed to determine the underlying constructs from the questionnaire (Alnaami et al., 2020; Sigudla and Maritz, 2023; Yu and Richardson, 2015). Descriptive statistics were also analyzed using mean and standard deviation (SD) to understand the demographic characteristics of ASHA workers (Pfeuffer et al., 2022; Shihundla et al., 2016).

The suitability of the data for factor analysis was verified using the Kaiser-Meyer-Olkin (KMO) test and Bartlett's Test of Sphericity, confirming the adequacy of the sample size and the presence of correlations among the variables. Factors were extracted using principal component analysis (PCA), with an eigenvalue as the criterion for factor retention. To enhance interpretability, varimax rotation was applied, leading to the identification of distinct factors representing different usability dimensions (Koopmann et al., 2023; Sigudla and Maritz, 2023; Yu and Richardson, 2015; Zhou et al., 2019; Taherdoost et al., 2014). To ensure reliability, internal consistency was tested using Cronbach's alpha (Alnaami et al., 2020; Shrestha, 2021; Sigudla and Maritz, 2023; Yu and Richardson, 2015). The analysis was performed using IBM SPSS Statistics, Version 29.

The queries analyzed from questionnaire were integrated with responses from informal interviews, task analysis, and field observations to ensure accuracy and reliability. Qualitative data complemented the quantitative findings (see Fig. 2), offering deeper insights into the usability challenges of the ASHA diary. The qualitative analysis followed a structured manual approach, where data from observations, informal interviews, and task analysis were transcribed and coded using a direct content analysis method. Codes were cross-compared to eliminate discrepancies and categorized into broader themes, while recurring patterns were analyzed for consistency and external validity, which is tabulated in Appendix B (Das and Dasgupta, 2015; Shihundla et al., 2016; Bist et al., 2023).

3. Results

3.1. Descriptive data

Table 1 shows the socio-demographic profile of ASHA workers. The workers' mean age ($\pm$ SD) was 40.17 ± 5.09 years. Maximum, 25 ASHA workers (43.9 %) belonged to the age group of 39–46 years, followed by 20 workers (35.1 %) in the age group of 32–39 years. The majority (43.9 %) of ASHAs were educated up to higher secondary school, and most of them, 23 (40.4 %), had experience in a range of 5–10 years. 52.6 % of ASHA workers lived in joint families, while 47.4 % lived in nuclear families. 52.6 % have pucca houses with proper space or rooms, and 31.6 % have semi-pucca houses categorized by space limitation. For example, some ASHA workers lived in homes with only 1 or 2 rooms, where they shared the space with their husbands and children and stored all their belongings. Apart from space limitations, the places where they usually live have poor lighting conditions. Thus, it's hard for them to organize their stuff properly. Moreover, a few (33.3 %) of them

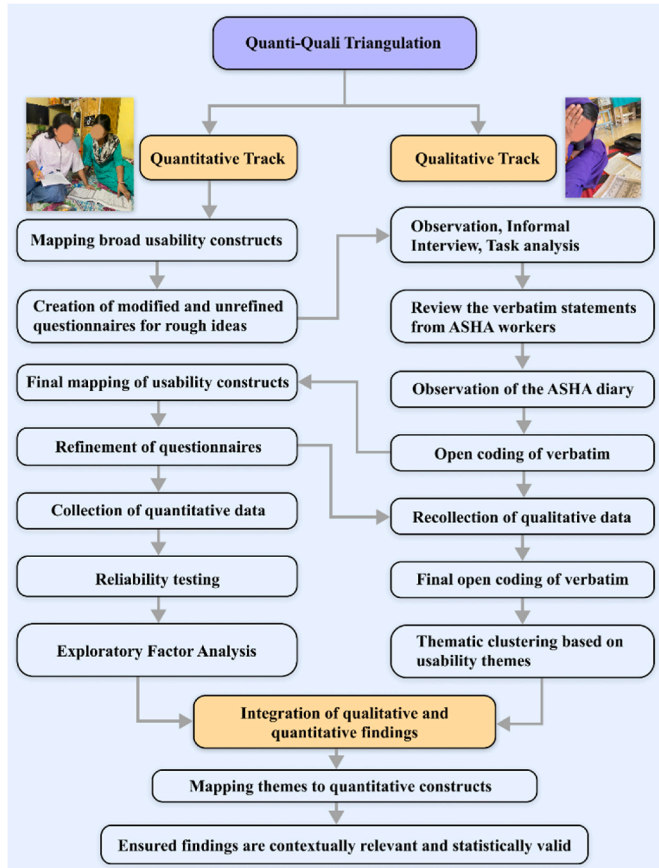


Fig. 2. Quanti-quali triangulation.

Table 1
Sociodemographic characteristics of ASHA workers.

Characteristics	Frequency	% [N = 57]
Age (in years)	Below 25 years	0
	25–32 years	4
	32–39 years	20
	39–46 years	25
	Above 46 years	8
	Mean (in years)	40.17
	Age range (in years)	29–51
Education	Primary (5th passed)	01
	Middle (8th passed)	14
	Higher Secondary (10th passed)	25
	Senior secondary (12th passed)	11
	Graduate (above 12)	06
	Less than 5 years	09
Experience	5–10 years	23
	10–15 years	18
	15–20 years	07
	More than 20 years	00
	Nuclear	27
Family type	Joint	30
	Kuccha	09
Type of house	Pucca	30
	Semi pucca	18
Number of children below 8th standards	0 child	38
	1 or 2 children	19
	≥3 children	00
Distance between home to workplace	0=<1 km	24
	1–5 km	32
	≥5 km	01
	Population covered	19
	1000–1199	22
	1200–1299	06
	≥1300	10
		17.5

had children who were under 8th standard, mostly dependent on their mothers. Most ASHA workers, 32 (56.1 %), travelled more than 1–5 km to cover the delegated population. Notably, 22 (38.6 %) of them were catering to a population of 1000–1199. The mean ($\pm$ SD) of the population catered 1089.75 ($\pm$ 364.93).

3.2. Sample adequacy and data suitability for factor analysis

A sample size of 57 ASHA workers was selected for this study. The size of the sample was a significant issue when conducting EFA. Several researchers recommended various guidelines on sample size (MacCallum et al., 1999; Sigudla and Maritz, 2023). For instance, Costello and Osborne (2005) suggested that smaller sample sizes can be justified in social science studies. Additionally, Appendix A presents the overall distribution of responses, including means and SDs.

KMO and Bartlett's test of Sphericity was initially conducted to examine the adequacy of the sample and the suitability of the data for factor analysis. KMO values range from 0 to 1, with values between 0.8 and 1.0 indicating adequate sampling, 0.7 and 0.79 indicating moderate sampling, and 0.6 and 0.69 indicating mediocre sampling. Values below 0.6 suggested inadequate sampling for factor analysis (Sigudla and Maritz, 2023). A KMO correlation value greater than 0.6 was considered acceptable for conducting EFA, especially for samples under 100 (MacCallum et al., 1999; Samuels, 2017). In this study, the KMO value was 0.608, which indicated that items in the dataset were strongly correlated with other items in the EFA correlation matrix, making the sample adequate for analyzing the EFA output (see Table 2). Bartlett's test of Sphericity (Bartlett, 1950) provided a chi-square output that must be significant to justify the use of factor analysis. The significant results indicated that the correlation matrix was not an identity matrix, ensuring that the variables were correlated enough for factor analysis. The degree of freedom (df) in Bartlett's test represents the number of

Table 2
Effectiveness of factor analysis.

Kaiser-Meyer-Olkin Measure of Sampling Adequacy		0.608
Bartlett's Test of Sphericity	Approx. Chi-Square (χ^2)	1803.55
	Degree of freedom (df)	666
	Significance (p)	<0.001

unique correlations being tested. A higher df suggests that more variables contribute to the correlation structure, making the test more robust. A p -value less than 0.05 is considered good, while a p -value less than 0.001 is considered best (Shrestha, 2021; Tavakol and Wetzel, 2020). The results presented in Table 2 show that Bartlett's test of Sphericity was significant ($\chi^2 = 1803.55$, $df = 666$, $p < 0.001$), confirming that the correlations among variables were sufficient to justify the use of factor analysis.

3.3. Extraction and retention of constructs: documentation and retrieval

PCA was used to extract the factors, as it is considered the best-performing data reduction approach while retaining most of the variance from the matrix (Costello and Osborne, 2005). For this study, the cumulative percent of variance extracted, Kaiser's criteria (eigenvalue >1 rule) (Kaiser, 1960), and the Scree test were the methods employed for retention of factors to determine the number of constructs to retain for rotation.

The eigenvalues greater than 1 are considered to draw distinctions between constructs. Table 3 depicted that these constructs have rotation sum of squared loadings accounting 84.66 % of the total variance, well above the acceptable threshold of 60 % (Sigudla and Maritz, 2023; Taherdoost et al., 2014). However, eleven constructs were identified, explaining about 85 % of the total variance, consistent with the eigenvalue >1 rule, ensuring that PCA effectively captured the main usability constructs in ASHA diary documentation. The scree plot in Fig. 3 further confirmed that these constructs account for most of the total variability in the data, while the remaining constructs contributed minimally, indicating they are less significant (Yu and Richardson, 2015). Therefore, selecting these eleven constructs is considered an optimal choice for analysis.

3.4. Rotated factor loadings: usability issues in ASHA diary documentation and retrieval

Varimax rotation was used to obtain the factor loadings, which were crucial for extracting the observed items associated with underlying constructs. The relationship between each item and the underlying constructs was described by the factor loading values. There was a known relationship between sample size and acceptable factor loading values. Factor loadings were considered "high" or "good" if they were 0.8 or greater, although achieving this level was rare in real data, particularly in social science studies. Typically, factor loadings in social

Table 3
Eigenvalues.

Component	Total	Percentage of Variance (%)	Cumulative Percentage (%)
1	6.40	17.29	17.29
2	5.77	15.61	32.89
3	3.95	10.69	43.58
4	2.76	7.47	51.05
5	2.63	7.10	58.15
6	2.47	6.68	64.82
7	1.75	4.72	69.54
8	1.63	4.42	73.96
9	1.46	3.96	77.92
10	1.42	3.84	81.76
11	1.08	2.91	84.66

* Extraction Method: Principal Component Analysis.

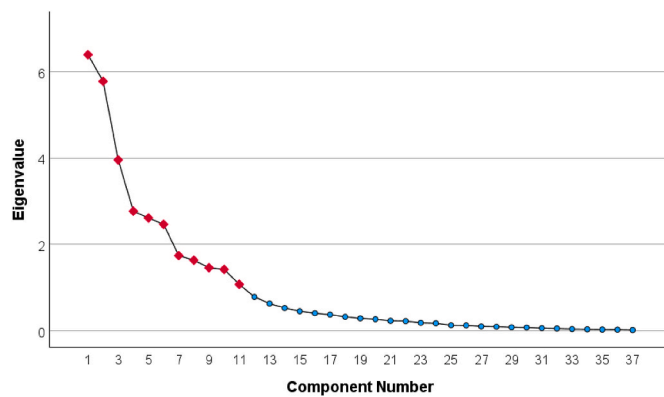


Fig. 3. Scree plot.

sciences range from low to moderate, with communalities between 0.40 and 0.70 (Costello and Osborne, 2005). MacCallum et al. (1999) stated that the detrimental effect of sample sizes was minimized when the communalities were consistently high, generally above 0.6. This research interpreted the rotational factor pattern using the criterion of communalities greater than 0.6. The items reflected the eleven constructs in the dataset, each having communalities greater than 0.6 after the extraction (see Table 4).

Table 4
Rotated component matrix.

Components	1	2	3	4	5	6	7	8	9	10	11
VC3	0.836										
VC5	0.820										
VC2	0.807										
VC1	0.796										
VC4	0.756										
S4		0.947									
S1		0.943									
S2		0.932									
S3		0.806									
CR3			0.918								
CR1			0.896								
CR4			0.879								
CR2			0.859								
EP4				0.910							
EP3				0.896							
EP1				0.823							
EP2				0.745							
FN1					0.816						
FN4					0.805						
FN3					0.802						
FN2					0.779						
CL2						0.924					
CL3						0.894					
CL1						0.794					
RU3							0.903				
RU1							0.834				
RU2							0.807				
Co2								0.878			
Co1								0.754			
Co3								0.723			
MM3									0.861		
MM1									0.854		
MM2									0.620		
Po1										0.928	
Po2										0.898	
IP1											0.896
IP2											0.870

*Loading values less than 0.6 have been removed, Components explained in Appendix A.

3.5. Reliability analysis for the construct: research characteristics

Cronbach's alpha values demonstrated the internal consistency reliability of constructs. The criteria for the interpretation of Cronbach's alpha coefficient for reliability were a value above 0.8 was a good reliability, in the range of 0.6–0.8 was an acceptable reliability, and a value below 0.6 was an unacceptable reliability for exploratory research (Koopmann et al., 2023; Sigudla and Maritz, 2023; Zhou et al., 2019). The analysis was conducted on each set of items for each construct in the questionnaire. Cronbach's alpha value for all eleven constructs is collectively 0.822 and is individually tabulated in Table 5, indicating

Table 5
Reliability analysis.

Constructs	Cronbach's alpha	Number of items
Visual Cues	0.898	5
Content Relevance	0.938	4
Comprehensibility (or Learnability)	0.856	3
Simplicity	0.930	4
Consistency	0.856	3
Findability (or Searchability or Complex Navigation)	0.866	4
Information Prioritization	0.852	2
Error Prevention	0.866	4
Minimal Memory Load	0.819	3
Resources utilization	0.864	3
Portability	0.915	2

good reliability of the constructs.

3.6. Qualitative findings

The findings highlighted various usability and environmental challenges ASHA workers faced in documenting and retrieving information. Field observations in rural areas revealed significant design flaws in the ASHA diary, leading to many diaries being partially or entirely empty as workers struggled to record data. However, those who were highly dedicated, had fewer household responsibilities, faced minimal social disturbances, and experienced no major health issues were more likely to complete the diary despite these challenges. The diary's complex and detailed format overwhelmed workers, making documentation burdensome. Additionally, the diary lacked quick and accurate information retrieval features, forcing workers to flip through multiple pages, increasing cognitive load and reducing overall efficiency. Environmental factors such as poor sitting postures, inadequate lighting, limited storage in the home, frequent travel, work in different weather, and poor infrastructure further hindered usability, particularly in remote areas.

Informal interviews highlighted missing key information, repetitive tasks, and workers relying on self-made registers instead of the ASHA diary. One of the ASHA workers stated: *"I first collect all the information in my self-made registers, which is easy to use, then forcibly transfer the information to the diary because the inspection officers only check the ASHA diary."* Many workers found the system misaligned with their workflow and comfort, leading to frustration, errors, and reluctance to complete entries. Task analysis revealed prolonged documentation times and difficulties in navigating and locating unclear labeling, formatting, and cross-reference entries. The lack of structured guidance makes the process tedious and prone to errors. Overall, the cumulative findings (see [Appendix B](#)) highlight the urgent need for a more user-friendly, well-structured, and accessible diary format that improves data documentation and retrieval.

4. Discussion

The design of the ASHA diary is pivotal for the effective work of ASHA workers and for improving communication within health systems, particularly in rural settings. This study examined the usability issues in the existing paper-based ASHA diaries that ASHA workers use to record patients' health information in rural areas. It identified key usability constructs, redeveloped them to capture broader issues, and validated the measurement instruments. This study also qualitatively investigated quantitative data to gain an accurate and deeper understanding. Comprehensive usability constructs are revealed through a modified questionnaire and pattern coding analysis. While there are challenges in generalizing the findings of this research, the study successfully demonstrated that the identified constructs possess strong psychometric properties and account for a significant amount of variance in the ASHA diary's effectiveness. As summarized in [Table 4](#), this research identified the usability problems associated with the ASHA diary and those experienced by ASHA workers. The key constructs influencing effective documentation and information retrieval include visual cues, content relevance, comprehensibility or learnability, simplicity, consistency, findability, information prioritization, error prevention, minimal memory load, resource utilization, and portability.

The visual cue construct encompasses visibility, readability, and legibility, and consists of five items with high reliability (Cronbach's alpha = 0.898, exceeding the 0.8 threshold). Factor loadings (0.79, 0.80, 0.83, 0.75, and 0.82) surpass the acceptable 0.6 threshold, confirming their strong contribution. This factor significantly influences usability, explaining 17.29 % of the total variance with an eigenvalue of 6.40. It is crucial to ensure the ASHA diary's elements remain visible under various conditions, with high contrast and appropriately sized text. Readability facilitates text processing, while legibility ensures clear character distinction through suitable fonts and adequate spacing,

aligning with prior research ([Lee and Kozar, 2012](#); [Oztekin et al., 2013](#); [Seffah et al., 2006](#)). Qualitative findings further support this, as one of the ASHA workers noted: *"I have to wear my spectacles before filling out the diary because the text is too small."*

Content relevance is another key construct, comprising four items. It focused on delivering accurate, up-to-date, and task-relevant information for community health workers. High reliability (Cronbach's alpha = 0.938) and strong factor loadings (0.918, 0.96, 0.879, 0.859) confirm its significance, explaining 10.69 % of the variance with an eigenvalue of 3.95 ([Lee and Kozar, 2012](#)). ASHA workers highlighted usability concerns, stating: (ASHA-1): *"There is only space to tick for vaccination completion, but no option to write the vaccination date. Since we require the information of date, I separately write it in my register."* ASHA-2 opined, *"I have noticed that some column in the diary is never used."* These insights underscore the need for design improvements, ensuring essential information is efficiently recorded while eliminating redundant or underutilized sections.

Comprehensibility (or learnability) accounts for 6.68 % of the variance (eigenvalue = 2.47) with high reliability (Cronbach's alpha = 0.856) and strong factor loadings (0.924, 0.894, 0.794) across three items. The findings highlight that clear instructions and simple language enhance understanding, reduce errors, and improve health intervention efficiency. Simplicity, another critical construct, explains 15.61 % of the variance (eigenvalue = 5.77). It comprises four highly reliable items (Cronbach's alpha = 0.930) with strong factor loadings (0.947, 0.943, 0.932, 0.806), indicating that a simple and concise interface minimizes unnecessary complexity, providing clear steps and facilitating efficient documentation. However, the diary was observed to be overly cluttered and text-heavy, potentially hindering usability. Consistency accounts for 4.42 % variance (eigenvalue = 1.63) with high reliability (Cronbach alpha = 0.856) and strong factor loadings (0.878, 0.754, 0.723). These findings indicate that maintaining consistency in structure, terminology, formatting, and language improves usability by reducing confusion, allowing ASHA workers to navigate the diary more easily and efficiently ([Lee and Kozar, 2012](#)).

Findability (or navigation) identified as a crucial usability factor, comprising four items (factor loading = 0.816, 0.805, 0.802, 0.779) with high reliability (Cronbach's alpha = 0.866). This construct explains 7.10 % of the variance (eigenvalue of 2.63), highlighting challenges in locating required information swiftly, especially in time-sensitive situations ([Lee and Kozar, 2012](#); [Seffah et al., 2006](#)). This is consistent with qualitative findings where one ASHA worker stated: *"I find the information I need by looking at headings from the title page and then searching manually,"* confirming that poor indexing hinders usability. Information prioritization explains 2.91 % of the variance (eigenvalue = 1.08) with high reliability (Cronbach's alpha = 0.852) and strong factor loadings (0.896, 0.870) across two items. However, the current information arrangement in the diary is inadequate, making data documentation and retrieval difficult ([Murphy et al., 2019](#)). Task analysis further confirmed navigation challenges, as workers struggled to locate relevant information efficiently.

Error prevention is a key construct, explaining 7.47 % of the variance (eigenvalue = 2.76), exhibits high reliability (Cronbach's alpha = 0.866) and strong factor loadings (0.910, 0.896, 0.823, 0.745). It highlights the need for clear instructions, and redundancy is essential to minimize errors. Qualitative findings reveal frequent mistakes due to complex layouts and repetitive documentation. One ASHA worker stated, *"I sometimes accidentally move from one section to another without realizing it."* A well-informed design can prevent such errors in health-care documentation. Additionally, the memory load is another concern, as high-effort tasks can cause frustration, impair communication, and delay decision-making, especially in stressful or demanding environments ([Alshehri and Alanazi, 2023](#); [Oztekin et al., 2013](#)). The construct explains 3.96 % of the variance (eigenvalue = 1.46) with high reliability (Cronbach's alpha = 0.819) and strong factor loadings (0.861, 0.854, 0.620) across three items, emphasizing the need for a simplified,

intuitive design to reduce cognitive burden.

Resource utilization, comprising three items, is another critical factor (Cronbach's alpha = 0.864, factor loadings = 0.903, 0.834, 0.807), explaining 4.72 % of the variance (eigenvalue = 1.75). This is consistent with qualitative findings where one ASHA worker stated, *"I use self-made registers because some columns, like the due-list, fill up within one or two months, making the diary unusable."* This highlights the need for a cost-effective, durable, flexible, and easily updatable diary for long-term usability and practicality. Additionally, poor lighting conditions negatively affected the documentation. Portability emerged as a significant usability construct, containing two items (factor loadings = 0.928, 0.898) with high reliability (Cronbach's alpha = 0.915). It accounts for 3.84 % of the variance (eigenvalue 1.42). The ASHA diary's weight, size, and bulky design demands greater physical effort, especially as ASHA workers travel long distances to visit patients by foot or bicycle (Seffah et al., 2006; Gore et al., 2022). The impracticality is heightened in constrained or crowded environments like small homes or community meetings. One worker stated: *"Carrying the diary while cycling to villages is difficult, so I sometimes leave it at home and note things down later."* This suggests that usability improvements should consider the diary's size and practical use to enhance portability.

Further, most ASHA diaries were found to be partially or completely empty, as many workers preferred using self-made registers. Regardless of their age, education, or experience level, they struggled with maintaining the diary due to its complex structure, time-consuming nature, and frequent errors. The data entry is challenging, especially when they are constantly on the move, visiting households, assisting in childbirth, managing immunizations, and responding to emergencies. It was also observed that some physically fit workers with fewer household responsibilities modified their diaries by simplifying entries to better suit their needs. This highlights the lack of adaptability in the official diary format, prompting workers to develop their own solutions for efficiency.

Poor working conditions at their workplaces, such as SHC and AWC, further hinder documentation. In many cases, workers are forced to sit on uneven floors and write in dimly lit crowded rooms, making it difficult to maintain neat and organized records. Additionally, diaries often suffer damage from dust, moisture, and poor storage, complicating data retrieval. These challenges impact the accuracy and completeness of health records, forcing ASHA workers to rely on memory or informal notes, leading to inconsistencies. The absence of a user-friendly, accessible, and durable documentation system remains a significant barrier to effective healthcare data management in rural India.

This study presents several practical and policy-relevant implications for improving the effectiveness of tools like the ASHA diary. A human-centered design approach is essential to enhance usability and ensure the diary aligns with the cognitive and operational capacities of ASHA workers (International Organization for Standardization, 2019). Trying harder will not work; instead, changing the system of care will. As Kruk et al. (2018) said, effective health systems in the 21st century must be people-centered to deliver high-quality, outcome-driven services. Redesigning the ASHA diary to reflect this principle is therefore imperative. The Ministry of Health and Family Welfare should adopt user-friendly design standards for all health documentation tools, which would facilitate easier data entry, enhance information retrieval, and support more accurate record-keeping at the community level.

Specific improvements such as simplifying the ASHA diary format, using visual cues and color coding, and aligning the structure with daily tasks can significantly reduce the documentation burden and improve data accuracy. Focused training on diary use and basic data aggregation can further minimize cognitive load and errors. Admon et al. (2013) suggested that if knowledgeable and capable ASHA workers are appointed as data assistants of peers, and further supervised by trained personnel at the PHC level, it can enhance the accuracy of monthly reporting, streamline data processes, and support overburdened and uncared frontline workers. Collectively, these measures can strengthen data quality and management while allowing ASHA workers to focus

more on patient care.

While this study provides valuable insights, it has some limitations. The sample size is small and limited to a specific geographic area, which may affect the generalizability of the findings. Additionally, a broader range of age groups should be covered to identify which groups are more influenced by usability problems. The study found very little work done to evaluate usability issues in the paper-based documentation process, so the criteria used may not be sufficient to identify all problems, especially for ASHA workers. However, the evidence generated by this research highlights current efforts to improve PPRs in LMICs. Future research should consider larger and more diverse samples and usability criteria to validate these results more accurately.

5. Conclusions

Improvement in documentation and information retrieval for ASHA workers cannot be done without enhancing the usability of the ASHA diary. The findings of this study underscore the critical role of a well-designed ASHA diary to augment the functionality and effectiveness of ASHA workers, particularly in rural healthcare settings. This study developed a comprehensive set of questionnaires for improving the usability of paper-based health records, ranging from visual cues and content relevance to portability and cognitive load. The rigorous methodological approach of the study, including the development and validation of robust usability constructs, not only illuminates the complexities of ASHA diary usage but also offers practical solutions to mitigate these challenges. The implications are far-reaching. It suggested that a human-centered redesign of the ASHA diary and assistantship could lead to significant improvements in documentation accuracy, efficiency, and overall user satisfaction. Such improvements are likely to directly impact the quality of healthcare delivery in rural areas, enabling ASHA workers to perform their duties more effectively and with less cognitive strain and workload. This research urges policymakers, designers, and healthcare professionals to prioritize usability in health documentation tools, ensuring ASHA workers are well-equipped to provide optimal care. The study's recommendations should guide future innovations, fostering more effective, user-friendly systems that support CHWs in LMICs.

CRedit authorship contribution statement

Sakshi Patel: Writing – original draft, Formal analysis, Validation, Data curation, Writing – review & editing, Methodology, Conceptualization. **Puneet Tandon:** Supervision, Conceptualization, Writing – review & editing.

Ethics approval

The evaluation was approved by the institutional review board at PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur, Madhya Pradesh, India. Moreover, oral approval from each participant was obtained before the data collection.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.socscimed.2025.118370>.

Data availability

The datasets collected and/or analyzed during the current work are available from the corresponding author upon reasonable request.

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Glossary

- ASHA (Accredited Social Health Activist):** Community health workers in India who bridge the gap between healthcare systems and rural populations, providing essential health services and documentation
- ASHA Diary:** An official record-keeping tool used by ASHA workers to document and retrieve community health-related data
- PPR (Paper Patient Record):** Physical documents used for manually recording and managing healthcare information
- Usability:** The ease and efficiency with which ASHA workers can use the ASHA diary to perform documentation and retrieval tasks accurately
- Exploratory Factor Analysis (EFA):** A statistical method is used to identify underlying relationships among observed variables, reducing them into meaningful constructs or factors without a predefined structure